

4 - Exploring and Expanding Agent-Based Modeling

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Course Outline (Modules)

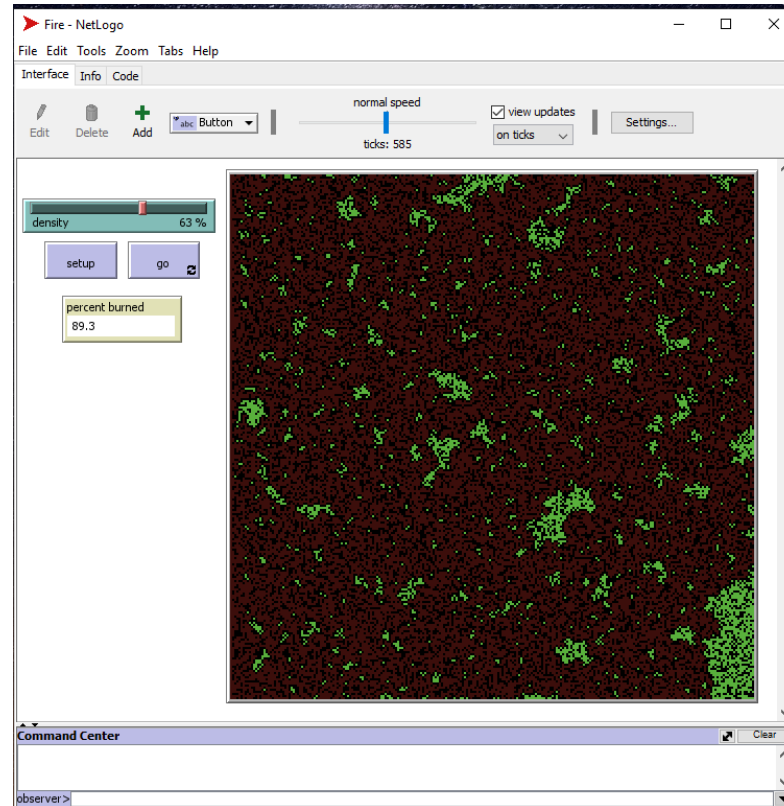
- Module 1. Introduction to Multi-Agent Systems
- Module 2. Intro to Agent-Based Modeling
- Module 3. On Using ABMs in Cognitive and Computational Social Science
- **Module 4. Exploring and Extending Agent-Based Modeling**
- Module 5. Creating Agent-Based Models
- Module 7. Adaptive Agents
- Module 8. Analyzing Agent-Based Models
- Module 9. More on Analysis of ABMs
- Module 10. Example publication combining ABM, MAS, and Cognition
- Module 11. Group Projects

Wooclap

Four characteristic features of ABM

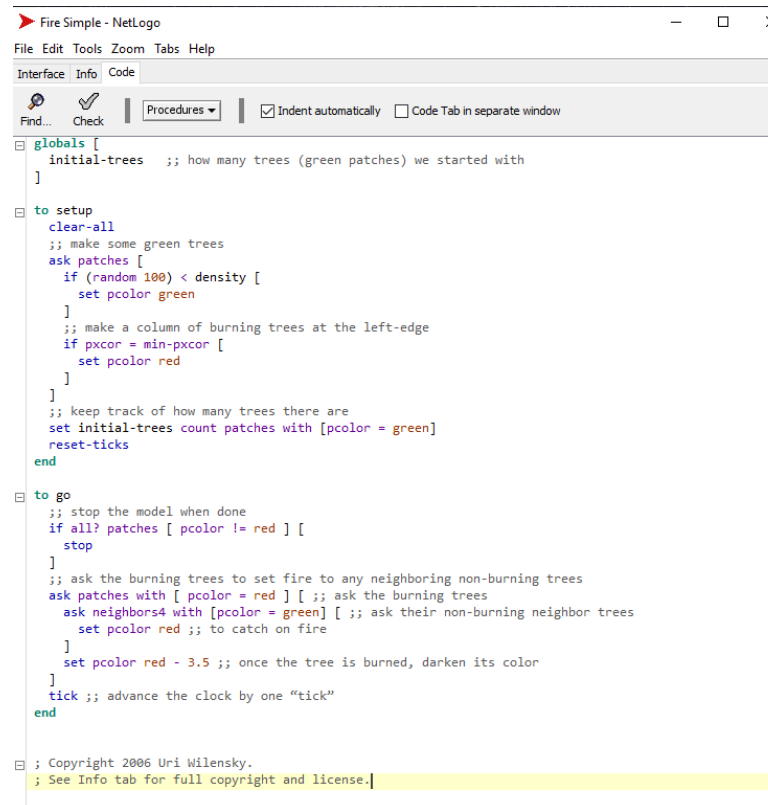
- Simple rules can generate complex phenomena
- Randomness in individual behavior can result in consistent patterns of population behavior
- Complex patterns can *self-organize* without any leader orchestrating the behavior
- Different models emphasize different aspects of the world

The Fire Model



Fire Model

The Fire Model



```
Fire Simple - NetLogo
File Edit Tools Zoom Tabs Help
Interface Info Code
Find... Check Procedures Indent automatically Code Tab in separate window

globals [
  initial-trees ;; how many trees (green patches) we started with
]

to setup
  clear-all
  ;; make some green trees
  ask patches [
    if (random 100) < density [
      set pcolor green
    ]
  ]
  ;; make a column of burning trees at the left-edge
  if pxcor = min-pxcor [
    set pcolor red
  ]
  ;; keep track of how many trees there are
  set initial-trees count patches with [pcolor = green]
  reset-ticks
end

to go
  ;; stop the model when done
  if all? patches [ pcolor != red ] [
    stop
  ]
  ;; ask the burning trees to set fire to any neighboring non-burning trees
  ask patches with [ pcolor = red ] [ ;; ask the burning trees
    ask neighbors4 with [pcolor = green] [ ;; ask their non-burning neighbor trees
      set pcolor red ;; to catch on fire
    ]
    set pcolor red - 3.5 ;; once the tree is burned, darken its color
  ]
  tick ;; advance the clock by one "tick"
end

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```

Fire Model

Fire: Probabilistic Transitions

- Fire spread isn't totally deterministic, but should be probabilistic
- Add a randomly generated number between 0-100 and if random number is less than probability of spread, the fire will spread to the next patch

Fire: Probabilistic Transitions

```
to go
  ;; stop the model when done
  if all? patches [ pcolor != red ] [ stop ]

  ;; ask the burning trees to set fire to any neighboring non-burning trees
  ask patches with [ pcolor = red ] [ ;; ask the burning trees
    ask neighbors4 with [ pcolor = green ] [ ;; ask their non-burning neighbor trees
      ;; only burn if a random draw is greater than the probability of spread
      if random 100 < probability-of-spread [ set pcolor red ] ;; to catch on fire
    ]
    set pcolor red - 3.5 ;; once the tree is burned, darken its color
  ]
  tick ;; advance the clock by one "tick"
end
```

Fire Model with Probabilities

Fire: Adding Wind

- Two sliders:
 - Wind from the south or north (with a negative wave)
 - Wind from the west or east (with a negative wave)
- Code that modifies the probability of spread depending on wind speed and direction

Fire: Adding Wind

```
to go
  if all? patches [pcolor = red]
    [ stop ]

  ;; each burning tree (red patch) checks its 4 neighbors.
  ;; If any are unburned trees (green patches), change their probability
  ;; of igniting based on the wind direction
  ask patches with [pcolor = red] [
    ;; ask the unburned trees neighboring the burning tree
    ask neighbors4 with [pcolor = green] [
      let probability probability-of-spread ;; define a local variable

      ;; compute the direction you (the green tree) are from the burning tree
      ;; (NOTE: "myself" is the burning tree (the red patch) that asked you
      ;; to execute commands)
      let direction towards myself
      ;; the burning tree is north of you
      ;; so the south wind impedes the fire spreading to you
      ;; so reduce the probability of spread
      if (direction = 0 ) [
        set probability probability - south-wind-speed ]

      ;; the burning tree is east of you
      ;; so the west wind impedes the fire spreading to you
      ;; so reduce the probability of spread
      if (direction = 90 ) [
        set probability probability - west-wind-speed ]

      ;; the burning tree is south of you
      ;; so the south wind aids the fire spreading to you
      ;; so increase the probability of spread
      if (direction = 180 ) [
        set probability probability + south-wind-speed ]

      ;; the burning tree is west of you
      ;; so the west wind aids the fire spreading to you
      ;; so increase the probability of spread
      if (direction = 270 ) [
        set probability probability + west-wind-speed ]
      if random 100 < probability [ set pcolor red ]
    ]
    set pcolor red - 3.5
  ]
end
```

Fire Model with Wind

Fire: Long-Distance Transmission

- Allow fire to jump over long distances without traveling from patch to patch
- Boolean switch for Big Jumps that move a number of patches ahead depending on the wind

Fire: Long-Distance Transmission

```
...
if random 100 < probability [
    set pcolor red

    ;; if big-jumps is on, then sparks can fly farther
    if big-jumps? [
        let target patch-at ( west-wind-speed / 5 ) ( south-
        wind-speed / 5 )
        if target != nobody and [pcolor] of target = green [
            ask target [
                set pcolor red ;; ignite the target patch
            ]
        ]
    ]
]
...
```

Fire Model with Wind

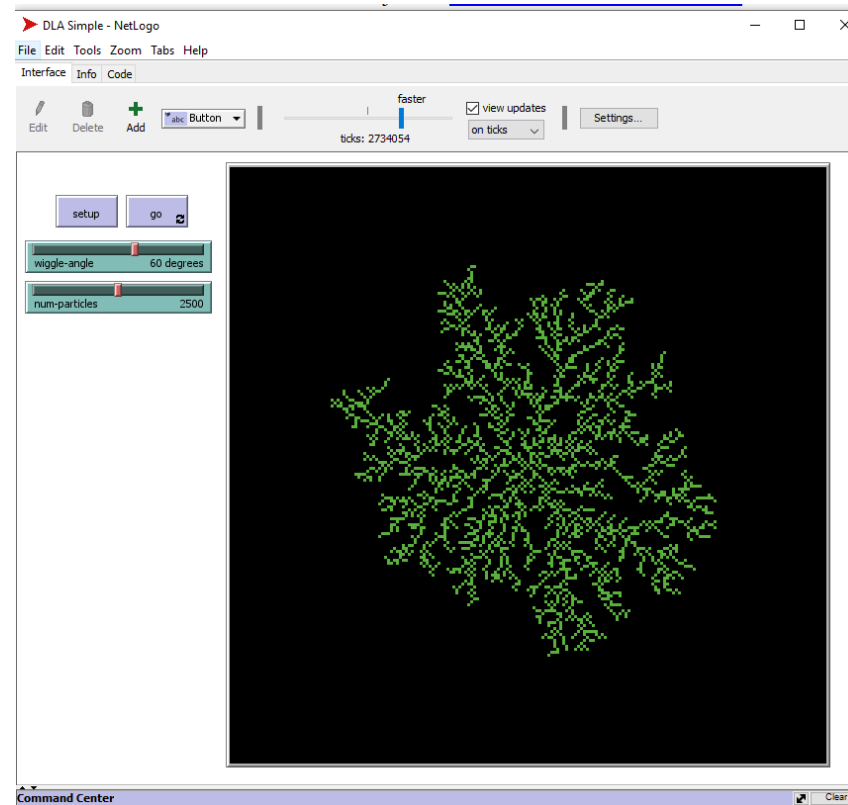
Exercise

- In the Fire Simple model, the fire is initialized on the left edge of the view. Modify it so that it starts at a single random location.
 - Can you redefine criticality under these conditions?
 - What is the critical threshold for spread under these conditions and with your definition of criticality?

Diffusion-Limited Aggregation (DLA) Model

- Idealization of the process of formation for crystals, coral, fungi, lightning, growth of lungs, and cities
- Particle agents wiggle (turn) and move around and 'dies' if any neighboring patches are green

Diffusion-Limited Aggregation (DLA) Model



DLA Model

Diffusion-Limited Aggregation (DLA) Model

```
to setup
  clear-all
  ;; start with one green "seed" patch at the center of the world
  ask patch 0 0 [
    set pcolor green
  ]
  create-turtles num-particles [
    set color red
    set size 1.5 ;; easier to see
    setxy random-xcor random-ycor
  ]
  reset-ticks
end

to go
  ask turtles [
    ;; turn a random amount right and left
    right random wiggle-angle
    left random wiggle-angle
    forward 1
    ;; if you are touching a green patch
    if any? neighbors with [ pcolor = green ] [
      set pcolor green ;; turn your own patch green
      die ;; and then die
    ]
  ]
  tick
end

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```

DLA Model

DLA: Probabilistic Sticking

- Change the probability that a particle agent stop if any neighbors are green (but also not on top of a green particle)
- Can generate 'thicker structures' (with $\text{prob} = 0.5$ and $\text{prob} = 1.0$ is original model)

DLA: Probabilistic Sticking

```
to go
  ask turtles
    [ right random wiggle-angle
      left random wiggle-angle
      forward 1
      if (pcolor = black) and ( any? neighbors with [pcolor = green] )
and ( random-float 1.0 < probability-of-sticking )
        [ set pcolor green
                                                die ] ]
  tick
end
```

DLA Prob Model Code

DLA: Neighbor Influence

- Make the particle more likely to stop if 2-3 neighboring patches are green compared to just

DLA: Neighbor Influence

```
to go
  ask turtles
  [ right random wiggle-angle
    left random wiggle-angle
    forward 1
    let local-prob probability-of-sticking

    ;; if neighbor-influence is TRUE then make the probability proportionate
    ;; to the number of green neighbors, otherwise use the slider as before
    if neighbor-influence? [
      ;; increase the probability of sticking the more green neighbors there are
      set local-prob probability-of-sticking *
      (count neighbors with [pcolor = green] / 8)
    ]

    if (pcolor = black) and ( any? neighbors with [pcolor = green] )
    and ( random-float 1.0 < local-prob )
    [ set pcolor green
      die ] ]
  tick
end
```

DLA Neighbors Influence Model

DLA: Different Aggregates

- Place a number of green patches in the world in random places
- We get multiple formations

DLA: Different Aggregates

```
to setup
  clear-all
  ;; start with NUM-SEEDS green patches as "seeds"
  ask n-of num-seeds patches [
    set pcolor green
  ]
  create-turtles num-particles [
    set color red
    set size 1.5 ;; easier to see
    setxy random-xcor random-ycor
  ]
  reset-ticks
end

to go
  ask turtles [
    right random wiggle-angle
    left random wiggle-angle
    forward 1
    ;; if neighbor-influence is TRUE then make the probability proportionate
    ;; to the number of green neighbors, otherwise use the slider as before
    let local-prob probability-of-sticking
    if neighbor-influence? [
      ;; increase the probability of sticking the more green neighbors there are
      set local-prob probability-of-sticking * (count neighbors with [ pcolor = green ] / 8)
    ]
    if any? neighbors with [ pcolor = green ]
      and pcolor = black
      and random-float 1.0 < local-prob [
        set pcolor green
        die
      ]
  ]
  tick
end
```

DLA Model With Aggregates

Exercise

- In the second extension of DLA Simple, we make the probability of sticking dependent on the number of neighbors. We accomplished this by multiplying the probability of sticking by the fraction of neighbors that are green.
 - Is this modeling choice realistic for DLAs?
 - **Propose and defend another way of modeling the changing probability.**

Segregation Model

- See chapter.

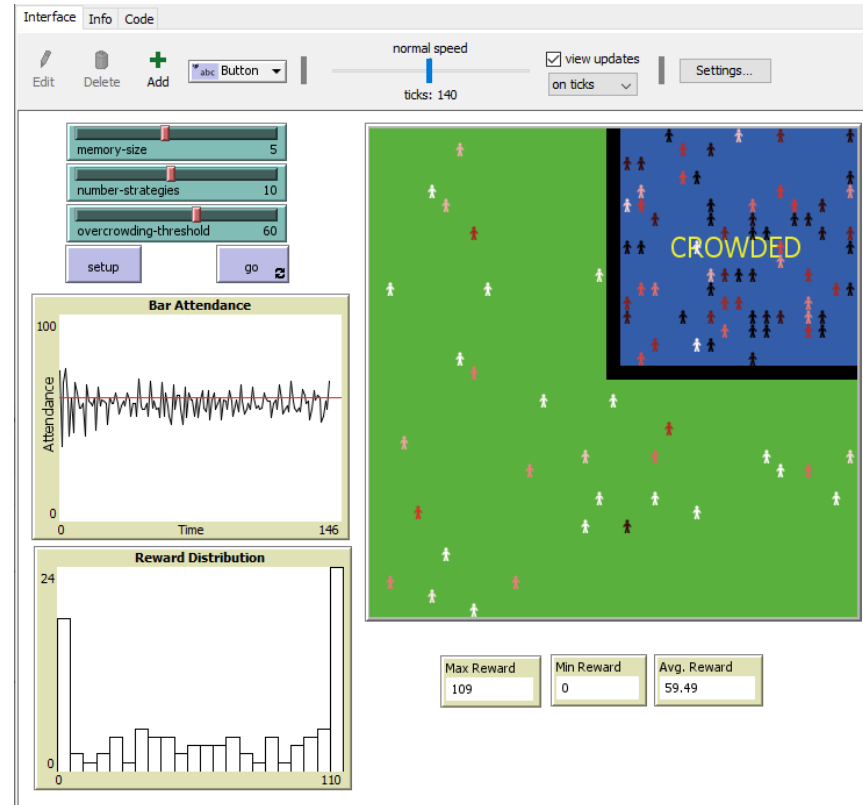
El Farol Model

- A model of bounded rationality and inductive reasoning (not completely rational agents maximizing utility)
- Agents don't perfectly optimize but an economic equilibrium is achieved
- How to decide to go to the bar to listen to Irish music, but you don't enjoy it when it's too crowded.
- Control agents memory for attendance in the prior weeks
- Strategies (rule's of thumb to use)

El Farol Model Extensions

- 1) Color agents that are better predictors of the bar's occupancy (agents get a reward when they visit and it isn't crowded). Dark agents with lowest reward.
- 2) Extract ave, min, max reward levels and display
- 3) Add histogram of reward values

El Farol Model Extensions



El Faro Model With Extensions

Exercise

- In the El Farol model, create displays for some other descriptive statistics that would be useful if the distribution of values were non-normal.
 - e.g., SD, variance,
 - [Link for Ideas](#)

Preparing for Module 5 - Creating Agent-Based Models

- Complete the reading:
 - CH4, Wilensky & Rand (2015)